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Ethics, Law and Professional Responsibility.

1. Differentiate between ethics, law and the professional responsibility with suitable examples from computing field.

Ans.

Aspect	Definition	Example in Computing
Ethics	Moral principles guiding behavior, not legally enforced.	A developer choosing not to collect unnecessary user data to respect privacy
Law	Legally enforceable rules, violation may lead to penalties.	GDPR compliance requiring user consent for data collection.
Professional Responsibility	Duties and standards expected by one's profession	Software engineers following ACM/IEEE Code of Ethics to ensure safe software design

Historical Ethical Failures in Technology.

2. Discuss the any two historical cases of ethical failures in technology and explain the lessons learned from them.

Ans. 1. Therac-25 Radiation Machine (1980s)

• The machine's software had software critical bugs causing massive overdoses of radiation, killing several patients.

• Lesson: Engineers must prioritize public safety over cost or deadlines, rigorously test safety-critical systems and take responsibilities for failures.

2. Cambridge Analytica/Facebook Data Scandal (2018)

• Millions of users' Facebook data were harvested without consent to influence political campaigns.

• Lesson: Transparency and user consent are very crucial, ethical data practices are as important as the legal compliance.

Software Engineering Ethics and Professional Responsibility.

3. Explain the principles of software engineering ethics, highlighting the issues related to professional

responsibility. Discuss how the ACM / IEEE Code of Ethics guides ethical decision-making in software engineering practices.

Ans. Principles of Software Engineering are:

1. Public safety: Engineers must avoid harm to users and society.

2. Honesty and Integrity: Report software capabilities and limitations truthfully.

3. Privacy: Respect confidentiality and secure sensitive information.

4. Competence: Only perform tasks you are qualified for, maintain continuous learning.

5. Professionalism: Accept responsibility for the decisions and outcomes.

ACM/IEEE Code of Ethics:

1. Provides guidance for ethical decision-making in software development.

2. Encourages reporting unsafe software,

protecting data privacy, and prioritizing social good over personal gain.

Example: If a developer finds a security flaw that could harm users, the code encourages prompt reporting and remediation even if it delays product release.

~~Utili Utilita~~

Utilitarianism vs Deontology in Computing

4. Compare and contrast Utilitarianism and Deontology (Kantian ethics) as ethical frameworks. How would you each approach handle a software privacy dilemma?

Ans.

Aspect	Utilitarianism	Deontology (Kantian ethics)
Definition	Ethical decisions are based on outcomes.	Ethical decisions are based on duty, rules and moral obligations
Privacy dilemma	May share anonymized user data for research	Cannot do data sharing without consent, regardless of potential benefits
Strength	Flexible, practical in large-scale decision-making	Consistent, protects individual rights
Weakness	Could harm some people for greater good	May ignore beneficial consequences.

Example: In AI healthcare, utilitarianism may allow sharing patient data to improve disease prediction, while deontology would require consent even if lives could be saved.

Virtue Ethics in Software Engineering.

5. Explain Virtue Ethics and describe how it can guide ethical decision-making for software engin...

Ans. • Virtue Ethics : Focuses on the character and moral values of an individual rather than rules or outcomes.

• Key virtues for engineers: Honesty, integrity, fairness, empathy and responsibility.

• Application in software engineering:

• Writing secure code, protecting user privacy and refusing shortcuts that could harm users.

• Example: A developer choosing to fix a serious security flaw rather than ignore it for a faster release demonstrates responsibility and integrity

Benefit: Encourages engineers to act ethically consistently, even when rules are unclear.

Explainability in High-Stakes AI systems

6. Why is explainability a key ethical requirement for AI systems used in high-stakes domains like credit lending or criminal justice? Discuss two

societal harms that can occur if an AI-system operates as a black box.

Ans. • Explainability ensures that AI decisions can be understood and justified by humans.

• Importance: Crucial in domains like credit scoring, hiring or criminal justice where the decisions have serious consequences.

Societal harms if AI is a black box:

1. Credit lending: Unexplained rejections may fairly deny loans, especially to marginalized communities.

2. Criminal justice: Black-box sentencing algorithms may perpetuate bias, leading to unjust convictions or harsher sentences.

Conclusion: Explainable AI is essential to maintain trust, fairness and accountability.

ACM Code of Ethics and its importance

7. What are the key principles of ACM code of Ethics? Why are such professional codes important for computing professionals?

Ans. Key principles:

1. Contribute to society and human well-being.
2. Avoid harm.
3. Be honest and trustworthy.
4. Respect privacy.
5. Honor intellectual property.
6. Maintain professional competence.

Importance:

- Provides guidance in complex, ambigious situations.
- Protects society and users from unethical practices.
- Builds trust in computing professionals.

Whistleblowing in Computing.

8. What is whistleblowing? Discuss the ethical considerations and challenges ~~a user~~ a computing professional might face when deciding to blow the whistle.

Ans. Reporting illegal, unsafe or unethical practices in an organization is called whistleblowing.

- Ethical Considerations
  - Duty to society vs loyalty to employer.
  - Potential retaliation or career loss.

- Example: Reporting unsafe software used in the medical devices to regulators.

Challenge: Engineers must carefully weigh the public safety against personal risk.

Conflict between Social Good and Privacy.

9. The ACM Code of Ethics emphasizes both contribute to society and human well-being and respect privacy. Describe a realistic scenario where these

two principles might come into conflict for a data scientist. How might the code guide a professional in resolving this tension?

Ans. Schenario: A data scientist at a public health department wants to use detailed smartphone location data to track disease spread and save lives. This directly conflicts with individual privacy rights.

Code Guidance:

• Weigh Principles: Recognize the validity of both contributing to society and respect privacy.

• Seek Least Harmful means (Avoid harm): Explore alternatives like anonymization, aggregation, differential privacy to achieve the public health goal while minimizing privacy intrusion.

• Transparency: Be open about the trade-offs.

The code pushes professional to find a creative solution that respects both principles to the greatest extent possible, rather than simply choosing one over the other.

10. Explain the difference between copyrights patents and trademarks in the context of software and digital products.

Ans.

Aspect	Copyright	Patents	Trademarks
Protect	Original expression	Non-obvious inventions.	Brand identifiers
Requirements	Originality	Novelty	Distinctiveness
Duration	+70/95 years	20 years	Perpetual
Scope	Narrow	Broad	Brand protection
Cost	Low	High, lengthy process	Moderate
Overlap	Code itself	Function behind the code	Brand of the product
Software example	Adobe Photo shop code	Amazon-1 click patent	Apple logo
Digital product	E-book text	MP3 compression method	Netflix name

11. Compare open-source licensing with proprietary software licensing. What are the ethical implications of each model.

Ans.

- Proprietary: Source code is secret. Users get a license to use the compiled software but cannot modify or share it.
  - Ethical Implications: Can incentivize innovation but restricts user freedom, autonomy and the ability to audit for security/privacy.
- Open-source: Source code is public. Users can use, study, modify and share it.
  - Ethical Implications: Promotes user autonomy, transparency and collaborative innovation. However, it raises ethical questions about the responsibility of maintainers to secure widely used, volunteer-driven components.

12. Define professional dissent and whistleblowing in computing. What are the key ethical considerations and potential risks a professional must weigh before deciding to blow the whistle on harmful or unethical practices within their organization?

Ans.

Professional Dissent: Internally challenging unethical practices through proper channels (e.g. report to a manager).

Whistleblowing: Exposing wrong doing externally after internal dissent fails or is not viable.

Key considerations before whistleblowing:

- Exhaust Internal Channels: Have you genuinely tried to fix it from within?
- Clarity and evidence: Is the wrongdoing clear and supported by evidence?
- Purity of Motive: Is the primary goal the public interest?

Risks:

- Severe personal stress and social ostracism.
- Legal retaliation from the employer.
- Possibility of being ineffective despite the personal cost.
- Professional and financial ruin (firing, blacklisting).

13. Discuss the concept of informed consent in personal data collection. What ethical principles should organizations follow when collecting user data?

Ans. Informed Consent: A user's agreement to data collection is only valid if it is:

1. Informed: User understands what data is collected, why, how it's used and who it's shared with.

2. Comprehended: Information is presented clearly - without jargon.

3. Voluntary: Consent is given freely, without coercion.

Ethical Principles for Organization:

i) Transparency: Be open and honest about data practices.

ii) Purpose Limitation: Only collect data for specified, legitimate reason.

iii) Data minimization: Only collect what is strictly necessary

iv) Security: Protect collected data as a trust.

v) User control: Provide easy ways to access, correct and delete data

15. Compare and contrast copyright and patents as forms of intellectual property protection in computing. What type of software asset is each best suited for and why?

Ans.

Feature	Copyright	Patents
Protects	The expression (the code itself)	The idea or process (invention)
Acquisition	Automatic upon creation	Formal application and examination
Cost	Low/Free	High (thousands in fees)
Duration	Very long (70+ years)	Short (20 years)
Infringement	Copying the code	Using the idea, even with different code

Best suited for:

- Copyright: Protecting the specific, literal code of any software project. It's the default, automatic protection.
- Patent: Protecting a novel, non-obvious functional innovation (e.g. a new compression algorithm). It provides a powerful ~~monolo~~ monopoly over the idea, blocking all competing implementations.

19. What are the ethical concerns associated with surveillance technologies? Explain with examples how these technologies can be misused.

Ans. Ethical Concerns:

i) Privacy violation: Erodes the private sphere and personal autonomy.

ii) Chilling effect: Discourages free expression and dissent.

iii) Power Assymmetry: Creates imbalance between watcher and watched.

iv) Bias: AI-driven surveillance can discriminate against minorities.

v) Function Creep: Technology deployed for one ~~person~~ purpose is used for another, more invasive one.

Example of Misuse:

• Government: Using "crime-fighting" facial recognition cameras to track and identify political protectors.

- Corporate: Using "bossware" to monitor employees' every keystroke and screen, creating a culture of distrust and enabling unfair penalization.

16. What is algorithmic bias? Explain its causes and suggest measures to ensure fairness in AI systems.

Ans. Systematic errors in AI that can create unfair outcomes, disadvantaging groups based on race, gender or socio-economic status.

Causes:

- Biased training data: Historical prejudices encoded (e.g hiring data favouring man).

- Sampling Bias: Unrepresentative data (e.g facial recognition trained mostly on light skin).

- Proxy variables: Natural data correlating with protected traits (e.g zip code for race).

Measures to Ensure Fairness:

- Diverse development teams.
- Dataset audits for representation.
- Fairness metrics (demographic parity, equal opportunity).
- Continuous monitoring after deployment.
- Impact assessments before launch.

17. Discuss the importance of transparency and explainability in AI. Who should be held accountable when an autonomous system causes harm?

Ans. Importance:

- Trust: Users must trust system safely.
- Debugging: Developers need to fix error.
- Justice: Affected individuals deserve the explanations.
- Compliance: GDPR mandates "right to explanation".

Accountables are when harm occurs:

- Developers: For coding errors or ignoring risks.
- Product Managers: For unsafe feature requirements.
- Companies: For deploying without oversight.
- Deployers: For using system in inappropriate contexts.

18. Discuss the role of ethical leadership in the IT organizations. How can leaders foster an ethical culture within their teams?

Ans. Role of Ethical Leaders:

Set organizational term, model integrity, prioritize values over short-term profits and create safe environments for raising concern.

Leaders foster Ethical culture by:

- Lead by example: Demonstrate integrity daily.
- Code of Conduct: Implement and enforce ethics codes.

• Safe channels: Anonymous reporting without retaliation.

• Reward Ethics: Recognize those raise concerns.

• Integrate into KPIs: Measure ethical performance alongside business metrics.

• Ethics Training: Regular workshops on dilemmas.

19. Differentiate between ethical hacking and cybercrime. Under what conditions is hacking considered ethical?

Ans.

Feature	Ethical Hacking	Cybercrime.
Purpose	Improves security, protect systems	Personal gain, damage, disruption
Authorization	Explicit permission obtained	No authorization
Outcome	Vulnerabilities fixed and reported	Theft, damage, extortion
Legality	Legal with consent	Illegal
Example	Penetration testing for a bank	Stealing credit card data

~~Et~~ Hacking is ethical:

- With explicit permission.
- Scope clearly defined
- Findings disclosed responsibly.
- No data theft or damage,
- Aim is to improve security.

20. Explain the concept of responsible disclosure in cybersecurity. What ethical guidelines should a security researcher follow when discovering a vulnerability?

Ans. A security researcher reports vulnerabilities privately to the organization so they can fix them before public disclosure, protecting users from the attackers.

Ethical Guidelines for Researchers:

1. Notify privately first: Contact organization directly, not public.

2. Allow Reasonable Time: Give 30-90 days for ~~[illegible]~~ fixing.

3. Avoid harm: Don't exploit beyond proof-of-concept

4. Provide details: Clear steps to reproduce.

5. Public disclosure only after fix; Or if the organization ignores critical flaw.

6. Follow bug bounty rules: If program exists, adhere to terms.

21. What is digital divide? Discuss its social implications and suggest measures to bridge this gap?

Ans. The gap between those who have access to digital technologies (internet, skills, devices) and those who do not is called digital divide.

Social Implications:

- Education Gap: Students without Internet fall behind.

- Economic Inequality: Limited access to jobs, online services.

• Healthcare disparities: Telehealth unavailable to marginalized groups.

• Reinforces poverty: Creates cycle of disadvantage.

Measures to bridge the gap:

i) Broadband connection in rural areas.

ii) Affordable device programs.

iii) Digital literacy training

iv) Public Wi-Fi in libraries, community centers.

v) Subsidized Internet for low-income households.

22. Explain the concept of green computing. How can computing professionals contribute to environmental sustainability?

Ans. Environmental responsible use of computing resources - designing, manufacturing, using and disposing of devices within ~~ea~~ minimal environmental impact

Computing Professionals can contribute through the following way:

i) Energy-efficient code: Optimize algorithm to reduce processing power.

ii) Cloud optimization: Right-size servers, auto-scaling to reduce waste.

iii) Virtualization: Reduce physical server footprint

iv) Remote Work: Enable policies that reduce commuting emissions.

v) Educate Users: Promote power-saving settings.

vi) E-waste reduction: Design for longetivity, repairability, recycling.

23. What is a conflict of Interest in the IT workspace? Provide examples and explain how professionals should ethically handle such situations.

Ans. A situation where a professional's personal interests could improperly influence their professional judgements on decisions.

Examples:

- Financial Interest: Recommending a vendor where you own stock.

- Side Hustles: Working for a competitor while employed full-time.

- Personal Relationships: Hiring a family member without disclosure.

- Self-Dealing: Using company resources for personal project

Ethical handling:

1. ~~Dist~~ Disclosure: Immediately reveal the conflict to HR/manager.

2. Recusal: Remove oneself from related decisions.

3. Company Policy: Follow organizational guidelines.

4. Documentation: Record the conflict and how it was handled.

5. When in Doubt: Disclose and ask - transparency is key

30. Explain how animal farm reflects the dangers of unchecked power and the moral responsibilities of leaders.

Ans. Animal Farm shows how power corrupts when it is not limited or monitored. The animals initially revolt to create a society based on equality and fairness ("All animals are equal"). Over the time the pigs especially Napoleon - gradually take the control of all decision-making

- Corruption of Power:
  - Napoleon eliminates his rival snowball to remove opposition
  - He uses fear (dogs) to silence criticism
  - Leadership becomes dictatorship instead of democracy.
  - Power shifts from serving others to controlling others.

- Political Manipulation:
  - Squealer spreads propaganda and distorts facts.
  - History is rewritten to justify Napoleon's action.
  - The commandments are secretly send to benefit the pigs.
- Betrayal of Equality:
  - Original belief: "All animals are equal."
  - Final version: "All animals are equal, but some animals are more equal than others."
  - This shows how leaders manipulate ideals to justify inequality.
  - Moral responsibilities of Leader:

Leaders should:

i) Act with honesty and transparency.
ii) Protect equality and justice.
iii) Be accountable for their reasons.
iv) Prioritize the welfare of the community.

31. Discuss the role of mentorship and personal growth in shaping Simba's ability to confront challenges and assume leadership.

Ans. The Ling King explores themes of courage, leadership, responsibility and moral growth. Simba's journey shows how mentorship shapes his development as a leader.

• Role of Mentorship:

i) Mufasa

• Teaches Simba about bravery and duty.

• Explains the "Circle of Life" and balance in nature.

• Provides moral foundation for leadership.

ii) Timon and Pumbaa:

• Offer emotional support and comfort

• Promotes "Hakuna Matata" (carefree life)

• Helps Simba heal but delay his acceptance of responsibility.

iii) Rafiki

- Encourages Simba to confront his past.
- Reminds him of his true identity.
- Inspires him to return and reclaim his role.

- Personal Growth

i) Simba runs away due to guilt and fear.

ii) Through guidance and reflection, he gains the maturity.

iii) Learns that leadership means facing challenges, not avoiding them.

iv) Returns to Pride Rock to restore justice and balance

- Leadership Lessons: A true leader must have:

i) Courage to face fear

ii) Responsibility towards the community

iii) Self-awareness.

iv) Moral commitment to restore harmony